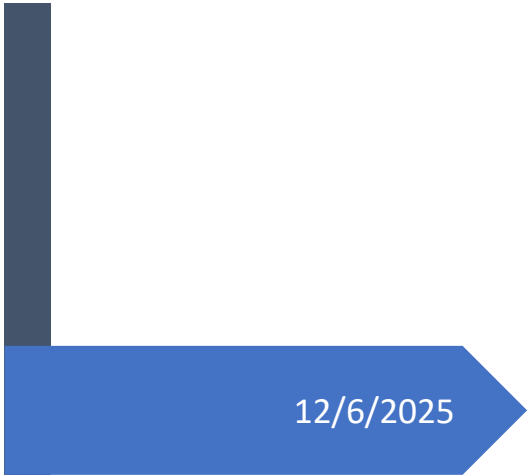


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12/6/2025

PROJECT REPORT

Project Title: “AI-Powered Customer Churn Prediction & Explainability System”

Submitted by: Vipul Sonawane

- **Live Web App (Streamlit):**
<https://telcoguard-churn-prediction-dc4vzjut8kxuuppszhhvqv.streamlit.app/>
- **GitHub Repository:**
<https://github.com/pruthvirajtarode/TelcoGuard-Churn-Prediction>

Abstract

Customer churn is one of the biggest challenges faced by telecom and service-based companies. Retaining existing customers is significantly more cost-effective than acquiring new ones.

TelcoGuard is an AI-powered churn prediction system that helps businesses:

- Predict which customers are likely to leave.
- Understand **why** they may leave using explainable AI (SHAP).
- Take proactive retention actions through a live web dashboard.

This system integrates **Machine Learning, Data Analytics, and Explainable AI (XAI)** and is deployed as a **real-time web application** using **Streamlit Cloud**.

Problem Statement

Telecom companies lose huge revenue due to unexpected customer churn. Traditional methods fail to:

- Predict churn early.
- Explain the reason behind churn.
- Provide actionable business insights.

Hence, there is a strong need for a **smart AI-based churn prediction system with explainability**.

Objectives

- Predict customer churn using Machine Learning
 - Provide real-time predictions via a web application
 - Explain predictions using SHAP
 - Visualize customer behavior trends
 - Enable business teams to take preventive actions
 - Deploy the system on cloud for real-time usage
-

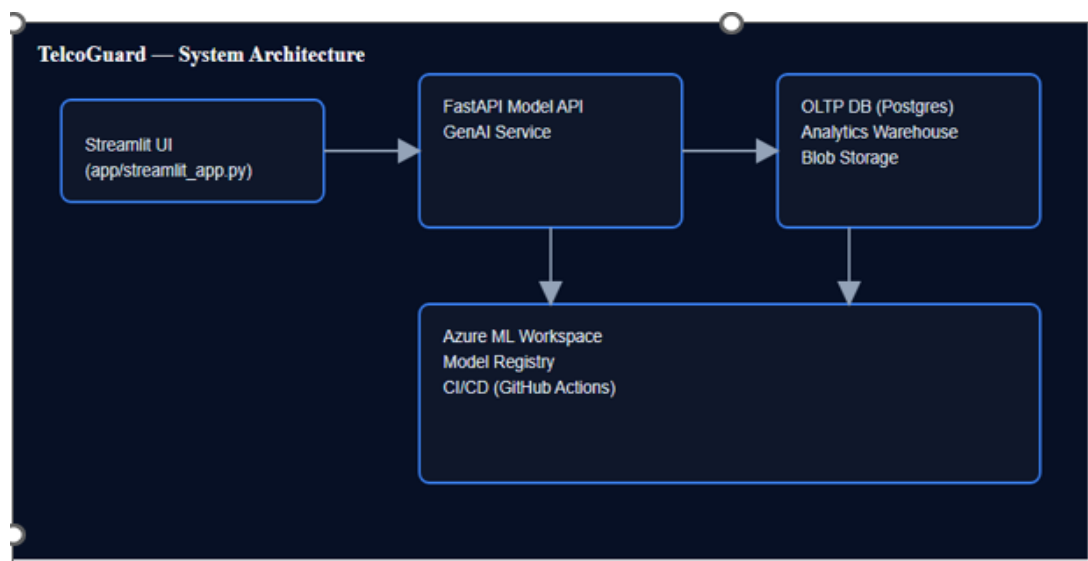
Technologies Used

Category	Tools
Programming	Python
ML Algorithms	Logistic Regression, Random Forest
Explainable AI	SHAP
Data Handling	Pandas, NumPy
Visualization	Matplotlib, Seaborn
Web Framework	Streamlit
Deployment	Streamlit Cloud
Version Control	Git & GitHub

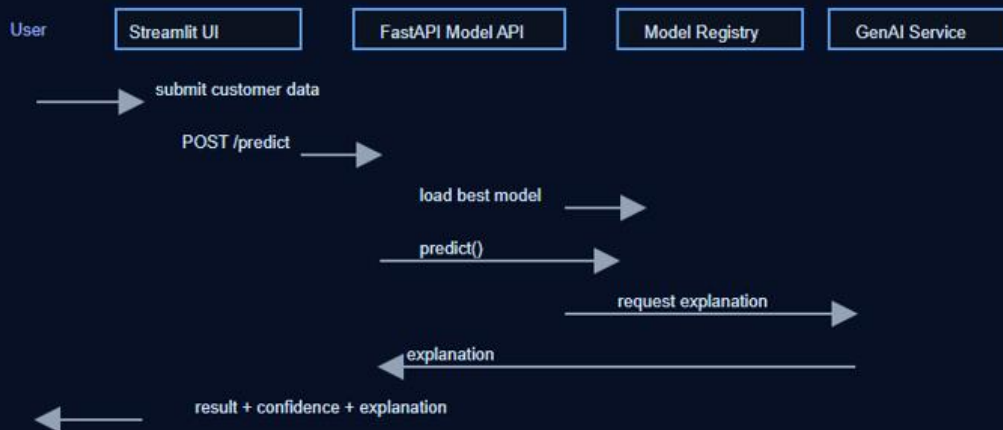
System Architecture (High-Level)

User → Streamlit Web App → ML Prediction Engine → SHAP Explainability → Results Dashboard

8. UML & System Diagrams (Add These Images in Report)



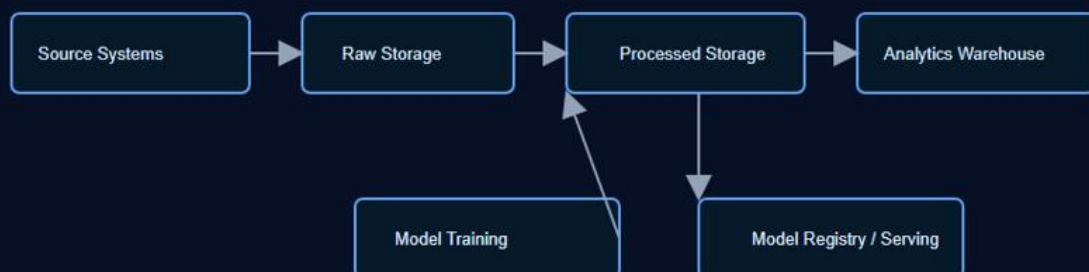
Prediction Flow — Sequence Diagram

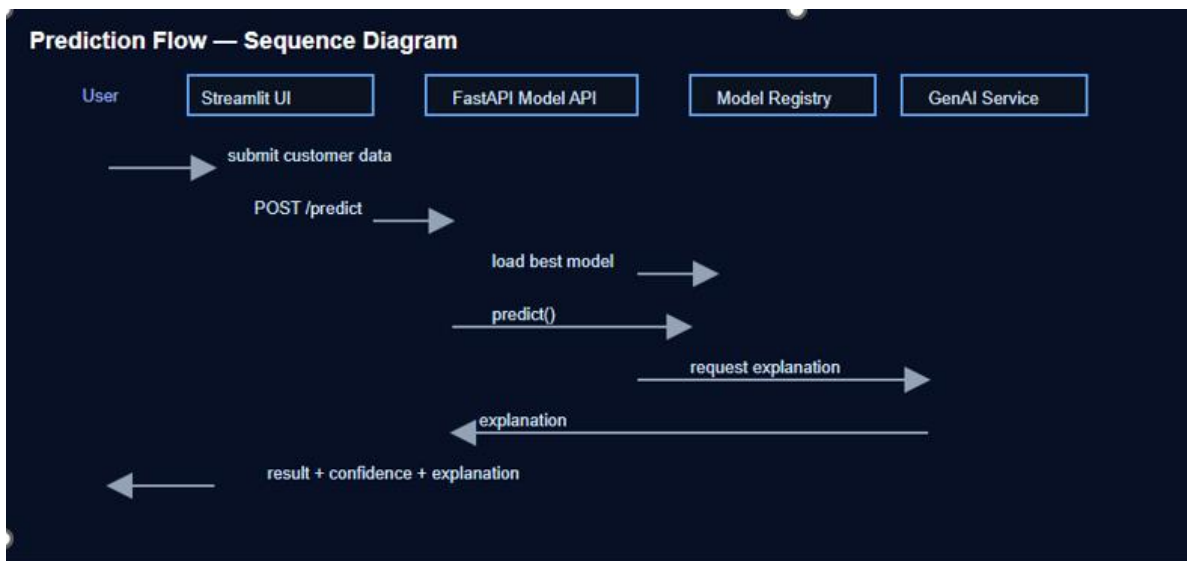
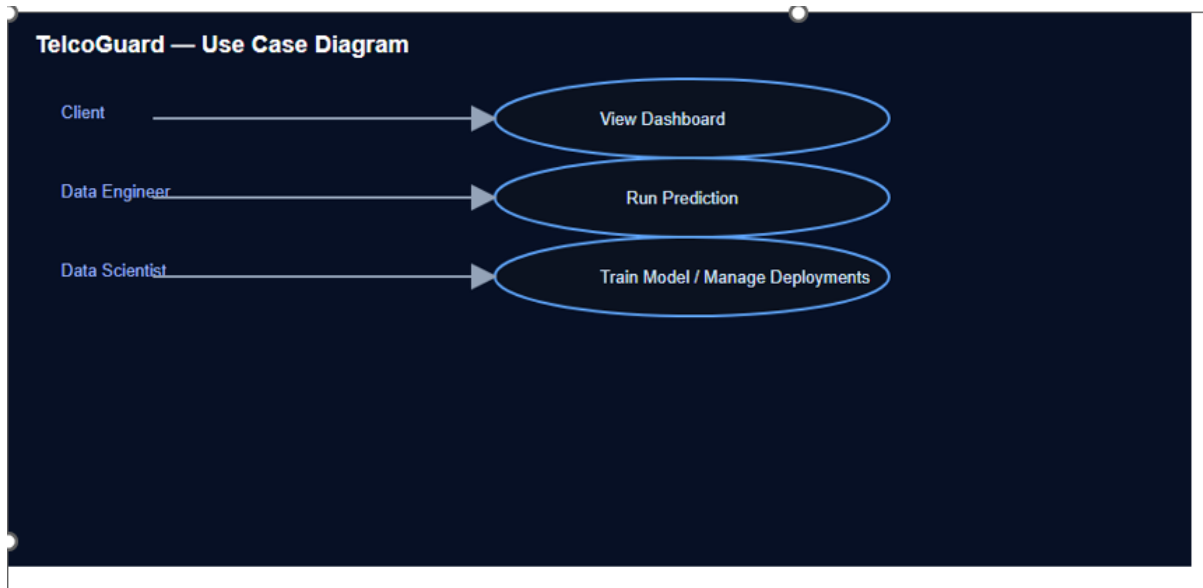


Deployment Diagram — TelcoGuard



Data Flow — TelcoGuard





9. Dataset Description

The system uses a **Telecom Customer Churn Dataset** containing:

- Customer demographics
- Subscription details
- Payment information
- Service usage metrics

- Churn status (Target Variable)

Target Output:

- 1 → Customer will churn
 - 0 → Customer will not churn
-

10. Machine Learning Model Details

- Data Preprocessing:
 - Missing value handling
 - Encoding categorical variables
 - Feature scaling
- Algorithms Used:
 - Logistic Regression
 - Random Forest Classifier
- Evaluation Metrics:
 - Accuracy
 - Precision
 - Recall
 - F1-Score

The model with the best performance is selected for deployment.

11. Explainable AI (SHAP Integration)

TelcoGuard integrates **SHAP (SHapley Additive exPlanations)** to:

- Show feature importance
- Explain individual customer predictions
- Improve business trust in AI decisions

SHAP outputs clearly show which factors contribute most to churn.

12. Web Application Features

- ✓ Customer data input
 - ✓ Live churn prediction
 - ✓ Probability score
 - ✓ SHAP explanation graph
 - ✓ Customer behavior visualizations
 - ✓ Business insights dashboard
 - ✓ Fully cloud deployed
-

13. Deployment Architecture

Client Browser → Streamlit Cloud → ML Model → SHAP Engine → Result Output

The system is fully accessible via the cloud with no local setup required for users.

14. Results & Performance

- Achieved high prediction accuracy
 - Real-time prediction speed
 - Visual interpretability using SHAP
 - Smooth UI experience for business users
-

15. Business Benefits

- ✓ Reduces customer churn
 - ✓ Improves customer retention strategies
 - ✓ Enhances business revenue
 - ✓ Builds trust through explainable AI
 - ✓ Supports data-driven decisions
-

16. Future Enhancements

- Deep Learning integration
- CRM system integration
- Automated retention recommendations

- API for mobile applications
 - Live database connection
-

17. Conclusion

TelcoGuard is a **powerful, real-time, cloud-deployed AI solution** that helps telecom and service companies:

- Predict churn accurately
- Understand customer behavior
- Make smarter retention decisions

This project successfully demonstrates:

- Machine Learning
- Explainable AI
- Data Visualization
- Cloud Deployment
- Real-World Industry Application